

AI Ecosystem for Fake Job Scam Detection and Fraud-Aware Job Recommendation

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Project Proposal Report

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DECLARATION

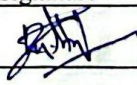
DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Online recruitment fraud has emerged as a significant cybersecurity threat in the modern labour market, with scam-based conversational fraud contributing to an estimated \$75 billion in global financial losses. While job seekers increasingly rely on platforms such as LinkedIn, ikman.lk, and TopJobs.lk, scammers exploit these platforms by publishing fraudulent job advertisements and subsequently moving communication to private messaging channels where they deploy psychological manipulation tactics to extract money and personal information from victims. Existing automated detection systems focus exclusively on fraudulent job advertisement classification and do not analyse the recruiter–candidate conversation phase where financial harm actually occurs. Furthermore, no existing system identifies the specific psychological manipulation tactic being deployed or provides actionable guidance for non-technical job seekers.

This research proposes the Behaviour-Inspired Semantic Tactic Detection (BSTD) framework — a mobile-based NLP system that detects psychological manipulation tactics in real-time recruiter–candidate conversations. A fine-tuned DistilBERT model performs simultaneous binary scam classification and multi-label tactic classification, identifying urgency pressure, FOMO, sunk-cost influence, and social proof — directly operationalising the theoretical behavioural findings of Anagha et al. into a working detection system. The system supports three input modes: a built-in chat platform, external message text input, and OCR-based screenshot analysis using OpenCV and Tesseract, enabling cross-platform detection without requiring API access from any messaging platform. Results are delivered through a React Native mobile application with a Flask REST API backend.

The system targets a minimum F1 score of 0.90 and a System Usability Scale score of 70 or above. This research contributes to SDG 8 (Decent Work and Economic Growth) and SDG 16 (Peace, Justice, and Strong Institutions), with commercial potential through integration with Sri Lankan employment platforms to protect job seekers at scale.

Keywords: Recruitment fraud, scam communication detection, NLP, DistilBERT, psychological manipulation tactics, BSTD

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
AUC-ROC	Area Under the Curve — Receiver Operating Characteristic
BERT	Bidirectional Encoder Representations from Transformers
BSTD	Behaviour-Inspired Semantic Tactic Detection
CCID	Computer Crimes Investigation Division
CoEAI	Centre of Excellence for Artificial Intelligence
DistilBERT	Distilled Bidirectional Encoder Representations from Transformers
DSR	Design Science Research
FOMO	Fear of Missing Out
FR	Functional Requirement
HTTPS	Hypertext Transfer Protocol Secure
ML	Machine Learning
NFR	Non-Functional Requirement
NLP	Natural Language Processing
OCR	Optical Character Recognition
OS	Operating System
PDPA	Personal Data Protection Act
REST	Representational State Transfer
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SMS	Short Message Service
SRS	Software Requirements Specification
SUS	System Usability Scale
TF-IDF	Term Frequency — Inverse Document Frequency
UI	User Interface

1.Introduction

1.1Background and Context

Online recruitment fraud has become a significant form of digital crime in the modern labour market. As job seekers increasingly rely on online platforms such as LinkedIn, ikman.lk, and TopJobs.lk to find employment, scammers exploit them to run recruitment scams. These scams typically follow a two-phase process. In the first phase, fraudsters publish deceptive job advertisements to attract potential victims. In the second phase, scammers move communication to private messaging channels such as WhatsApp, Telegram, SMS, or email, where they attempt to obtain money, personal documents, or banking information from victims.

Recent studies highlight the scale and structure of this problem. Tyszkiewicz and Noor [15] report that scam-based conversational fraud has contributed to global financial losses estimated at \$75 billion. Pitumpe and Rahmati [1] examined 29,209 fraudulent messages from 7,028 scammer contact points on SMS, WhatsApp, and Telegram, revealing that scammers recycle message templates and engage in structured communication patterns across platforms. In the Sri Lankan context, the Computer Crimes Investigation Division (CCID) has reported a consistent rise in online employment scam complaints in recent years, particularly affecting young job seekers and recent graduates. Behavioral research by Anagha et al. [2] identified four psychological manipulation tactics frequently used in scam conversations: urgency pressure, fear of missing out (FOMO), social proof, and sunk-cost influence.

However, existing automated detection systems mainly analyze fraudulent job advertisements and do not examine recruiter–candidate conversations where financial harm occurs. In addition, these systems typically provide only a binary classification of messages without identifying the manipulation tactics used by scammers. To address this gap, this research proposes Behaviour-Inspired Semantic Tactic Detection (BSTD), an NLP-based approach that detects psychological manipulation tactics in recruitment scam conversations and provides explainable warnings to help job seekers avoid fraud. This research contributes to SDG 8 (Decent Work and Economic Growth) and SDG 16 (Peace, Justice, and Strong Institutions). This work is conducted under CoEAI — the Centre of Excellence for Artificial Intelligence at SLIIT.

1.2 Literature Review

A review of recent scholarly literature indicates that online recruitment fraud has evolved into a sophisticated, multi-stage cybersecurity threat. Existing research has explored machine learning, NLP, and behavioural analysis to detect fraudulent job advertisements and malicious communications. However, no existing system adequately addresses the recruiter–candidate conversation stage where financial loss from recruitment fraud typically occurs.

1.2.1 Existing Research Approaches

Fake Job Post Detection

Alghamdi and Alharby [1] applied SVM and Random Forest classifiers to the EMSCAD dataset, achieving 97.41% accuracy in detecting fraudulent job advertisements, establishing it as the widely adopted benchmark for recruitment fraud research. Ullah and Jamjoom [2] extended this with an ensemble of AdaBoost, XGBoost, and Random Forest models, demonstrating strong advertisement-level classification performance. Taneja et al. [3] proposed Fraud-BERT, achieving F1 of 0.93 on EMSCAD using a fine-tuned transformer model. Rabai et al. [4] confirmed that deep learning consistently outperforms traditional ML on imbalanced fraud datasets, while Amaar et al. [5] demonstrated that NLP-based feature extraction improves detection over structured field classifiers. However, all five approaches analyse only static job advertisement content and cannot process recruiter–candidate conversational messages where financial harm actually occurs.

Scam Message Detection

Jain et al. [6] validated BERT-based models for SMS smishing detection, confirming that transformer architectures capture contextual patterns more effectively than keyword-based classifiers. Ghourabi [7] proposed SM-Detector, a BERT-based model for fraudulent SMS detection in mobile environments. Teja et al. [8] applied machine learning to WhatsApp chat data, demonstrating that conversation-level analysis is technically feasible. However, all three systems are restricted to a single platform and produce only binary scam classifications without identifying manipulation tactics. Songailaitė et al. [9] found that DistilBERT achieves comparable performance to larger transformer models at significantly lower computational cost, directly justifying its selection for real-time mobile deployment in this research. Uddin et al. [10] combined transformer models with attribution-based explainability for phishing detection, though their outputs target security researchers rather than non-technical users.

OCR-Based Detection

Shinde et al. [11] demonstrated that OCR can extract scam text from screenshot images without requiring platform API access, validating the conceptual foundation of the screenshot analysis mode in this research. However, their post-OCR classification used traditional models including K-means and Bi-LSTM, which lack the contextual understanding needed to detect paraphrased scam expressions.

Behavioural Analysis

Pitumpe and Rahmati [12] analysed 29,209 scam messages from 7,028 scammer contact points across SMS, WhatsApp, and Telegram, confirming that scammers systematically reuse message templates across platforms. Anagha et al. [13] identified four psychological manipulation tactics consistently deployed in recruitment scams — urgency pressure, FOMO, sunk-cost influence, and social proof — providing the theoretical foundation for the BSTD framework proposed in this research. However, these findings have not yet been integrated into any automated real-time detection system. Mahbub et al. [14] demonstrated that contextual employer signals further improve fraud detection accuracy when combined with textual analysis.

1.2.2 Existing Products and Services

Employment platforms such as LinkedIn and ikman.lk use internal moderation systems to detect suspicious job advertisements before publication. Mobile platforms provide limited sender-based filtering against known suspicious contacts. SafeChat [15] is the most relevant research prototype, performing conversation-level scam detection by assigning cumulative risk scores to SMS conversations. However, SafeChat targets general scam conversations rather than recruitment fraud, is restricted to SMS only, and does not identify which psychological manipulation tactic is being deployed. No publicly available system currently provides real-time tactic detection within recruitment conversations across multiple platforms.

1.2.3 Limitations of Existing Systems

Phase-1 Focus: All reviewed systems analyse job advertisements only and provide no protection once communication moves to private messaging channels where financial harm occurs.

Binary Classification: Existing systems classify messages as scam or legitimate without identifying the specific psychological tactic being used.

Stateless Analysis: Current systems analyse individual messages in isolation and cannot detect how manipulation strategies escalate across a full conversation.

Limited Explainability: Technical outputs and risk scores are not interpretable by non-technical job seekers, limiting practical usefulness.

These limitations collectively confirm the need for a system that goes beyond job advertisement detection and analyses the full recruiter–candidate conversation where financial harm actually occurs. The proposed **Behaviour-Inspired Semantic Tactic Detection (BSTD) framework** addresses all four limitations by analysing conversation-level messages across multiple platforms, identifying which specific psychological manipulation tactic the scammer is deploying, and providing plain-language explainable warnings that job seekers can immediately understand and act upon — translating the theoretical behavioural findings of Anagha et al. [2] into a working real-time detection system for the first time.

1.3 Research Gap

The review of existing literature exposes four critical research gaps that this study addresses.

First, most existing detection systems focus on identifying fraudulent job advertisements and do not analyse recruiter–candidate conversations where financial harm actually occurs. Systems such as Fraud-BERT [3], Alghamdi and Alharby [4], and Ullah and Jamjoom [5] are all confined to static job posting text and provide no protection once communication moves to private messaging channels.

Second, existing scam message detection systems including Jain et al. [9], Ghourabi [13], and Teja et al. [8] are each restricted to a single platform — SMS or WhatsApp only — and cannot operate across multiple messaging channels simultaneously, leaving victims significantly exposed.

Third, most existing systems produce only a binary classification of scam or legitimate without identifying which psychological manipulation tactic the scammer is deploying. Behavioural research by Anagha et al. [2] identified that scammers frequently employ urgency pressure, FOMO, sunk-cost influence, and social proof when manipulating victims. However, their work presents a theoretical analysis rather than an automated detection framework. To the best of the author's knowledge, these behavioural insights have not yet been operationalised into computable real-time detection signals within any NLP-based system.

Fourth, Shinde et al. [10] demonstrated that OCR can extract scam text from screenshot images without platform API access, but their post-OCR classification used traditional models lacking transformer-level contextual understanding.

This research addresses all four gaps through the proposed **Behaviour-Inspired Semantic Tactic Detection (BSTD) framework**, which analyses recruiter–candidate conversations, operates across multiple platforms via text input and OCR-based screenshot analysis, and translates Anagha et al.'s [2] theoretical behavioural findings into computable real-time tactic detection signals with plain-language explainable warnings for job seekers.

2. Objectives

2.1 Main Objective

To design, develop, and evaluate a Behaviour-Inspired Semantic Tactic Detection (BSTD) system that analyses recruiter–candidate conversation messages using NLP techniques, detects psychological manipulation tactics in real time, and delivers explainable warnings to job seekers — achieving a minimum F1 score of 0.90 and a System Usability Scale (SUS) score of 70 or above by the end of the academic year.

2.2 Specific Objective

1. To review and critically analyse existing literature on recruitment fraud detection, scam communication analysis, and psychological manipulation tactics in order to identify the research gaps addressed by this study.
2. To construct and annotate a dataset of recruiter–candidate scam conversation messages collected across multiple platforms including WhatsApp, Telegram, SMS, LinkedIn, and email, with labels for both binary scam classification and psychological manipulation tactic type.
3. To implement a baseline scam message classifier using TF-IDF feature extraction and Logistic Regression, evaluate its performance on the annotated dataset, and establish a measurable performance benchmark that demonstrates the improvement achieved by the deep learning approach.
4. To fine-tune a DistilBERT-based multi-label classifier on the annotated scam conversation dataset to detect and identify which of the four psychological manipulation tactics — urgency pressure, FOMO, sunk-cost influence, and social proof — is present in a given message, directly operationalising the theoretical behavioural findings of Anagha et al. [2] into a working NLP detection system for the first time.
5. To develop an OCR-based screenshot analysis pipeline that extracts conversation text from screenshot images without requiring platform API access, enabling cross-platform scam detection across WhatsApp, Telegram, SMS, LinkedIn, and email.
6. To design and deploy a mobile application with a Flask REST API backend that delivers real-time scam classification, tactic identification, and plain-language explainable warnings to job seekers, evaluated through a user study achieving a SUS score of 70 or above.

3. Methodology

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, appropriate for studies that produce and evaluate an IT artefact addressing a real-world problem. The research proceeds through five iterative phases: problem identification, solution design, artefact development, evaluation, and communication.

3.2 Technical Methods and Algorithms

Dataset Construction

A custom dataset of recruiter–candidate scam conversation messages will be constructed by collecting real scam conversations from participants who have previously encountered recruitment fraud, across WhatsApp, Telegram, SMS, LinkedIn, and email. The Anansi dataset [1] will also be used as an additional source, subject to access approval from the authors. Each message will receive a binary label (scam or legitimate) and a tactic label (urgency pressure, FOMO, sunk-cost influence, social proof, or none) based on the behavioural taxonomy of Anagha et al. [2]. Two independent annotators will label a stratified sample to measure inter-annotator agreement using Cohen's Kappa, targeting $\kappa \geq 0.80$. The final dataset will be split 80/10/10 for training, validation, and testing.

Baseline Model — TF-IDF and Logistic Regression

A TF-IDF feature extractor combined with Logistic Regression will be implemented as the performance baseline using unigram and bigram features. This establishes a measurable benchmark against which DistilBERT improvements will be directly compared.

Main Model — Fine-Tuned DistilBERT

DistilBERT (distilbert-base-uncased) will be fine-tuned using HuggingFace Transformers for two tasks: binary scam classification and multi-label tactic classification identifying which of the four psychological manipulation tactics from Anagha et al. [2] is present in a given message. This directly operationalises their theoretical behavioural findings into a working NLP detection system. Training parameters: learning rate 2×10^{-5} , batch size 16, AdamW optimiser, early stopping with patience of 3 epochs.

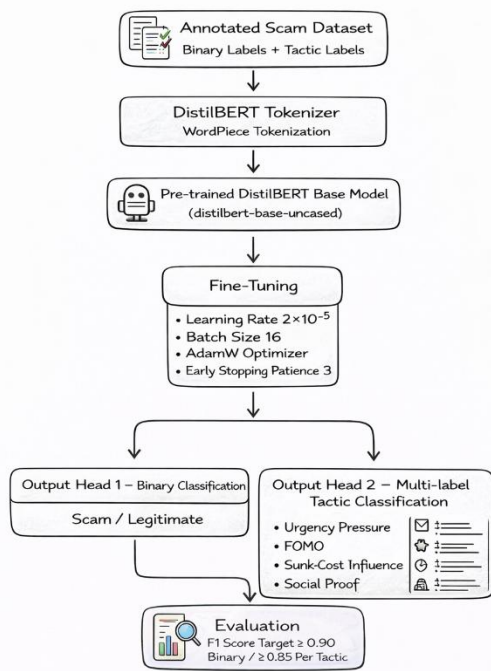


Figure: BSTD DistilBERT Fine-Tuning Pipeline

Figure 1: DistilBERT Fine-Tuning Pipeline

OCR Screenshot Pipeline

OpenCV will handle image pre-processing including denoising and adaptive thresholding to improve text clarity. Tesseract (pytesseract) will then extract conversation text from the processed screenshot images without requiring platform API access. Extracted text is passed directly into the DistilBERT classification pipeline, enabling cross-platform detection across WhatsApp, Telegram, SMS, LinkedIn, and email.

Explainability Layer

Each classification output will generate a plain-language warning mapped to the detected tactic. For example, a message classified as urgency pressure triggers: *"This recruiter is creating artificial time pressure. Legitimate employers do not impose same-day deadlines"*

3.3 Tools and Platforms

Table 1: Tools and Platforms

Tool / Platform	Purpose	Justification
Python 3.11	ML development	Industry standard for NLP/ML
Hugging Face Transformers	Distil BERT fine-tuning	Best transformer library
scikit-learn	TF-IDF baseline	Standard ML library
React Native	Mobile app (iOS & Android)	Cross-platform, single codebase
FastAPI	REST API backend	Lightweight, Python-native
OpenCV	Image pre-processing	Best open-source image processing
Tesseract (pytesseract)	OCR text extraction from screenshots	Free, open-source, offline capable
MongoDB Atlas	Data storage	Schema-flexible NoSQL for varied job structures
GitHub Actions	CI/CD pipeline	Automated testing and deployment

3.4 Data Collection and Ethical Considerations

A custom dataset will be constructed by collecting real recruiter–candidate scam conversation messages from participants who have previously encountered recruitment fraud, across WhatsApp, Telegram, SMS, LinkedIn, and email. The Anansi dataset [1] will also be used as an additional source, subject to access approval from the authors. Participant data will be collected only after written informed consent is obtained and participation will be voluntary. All messages will be immediately anonymised — personal identifiers replaced with synthetic ID codes before any researcher accesses the content. No message content will be stored on the server after classification results are returned.

This study will be conducted following approval from the SLIIT Research Ethics Committee and all procedures will comply with the Personal Data Protection Act (PDPA) No. 9 of 2022 of Sri Lanka.

3.5 Validation Strategy and Performance Metrics

Model performance will be evaluated using a stratified 80/20 train-test split. Primary metrics include Precision, Recall, F1 Score, and AUC-ROC. This research aims to achieve a target F1 score of 0.90 for binary scam classification and 0.85 per tactic class. The TF-IDF baseline and DistilBERT model will be directly compared to demonstrate measurable improvement. System usability will be evaluated through a user study using the System Usability Scale (SUS), aiming for a score of 70 or above.

4. High-Level Architecture Diagram

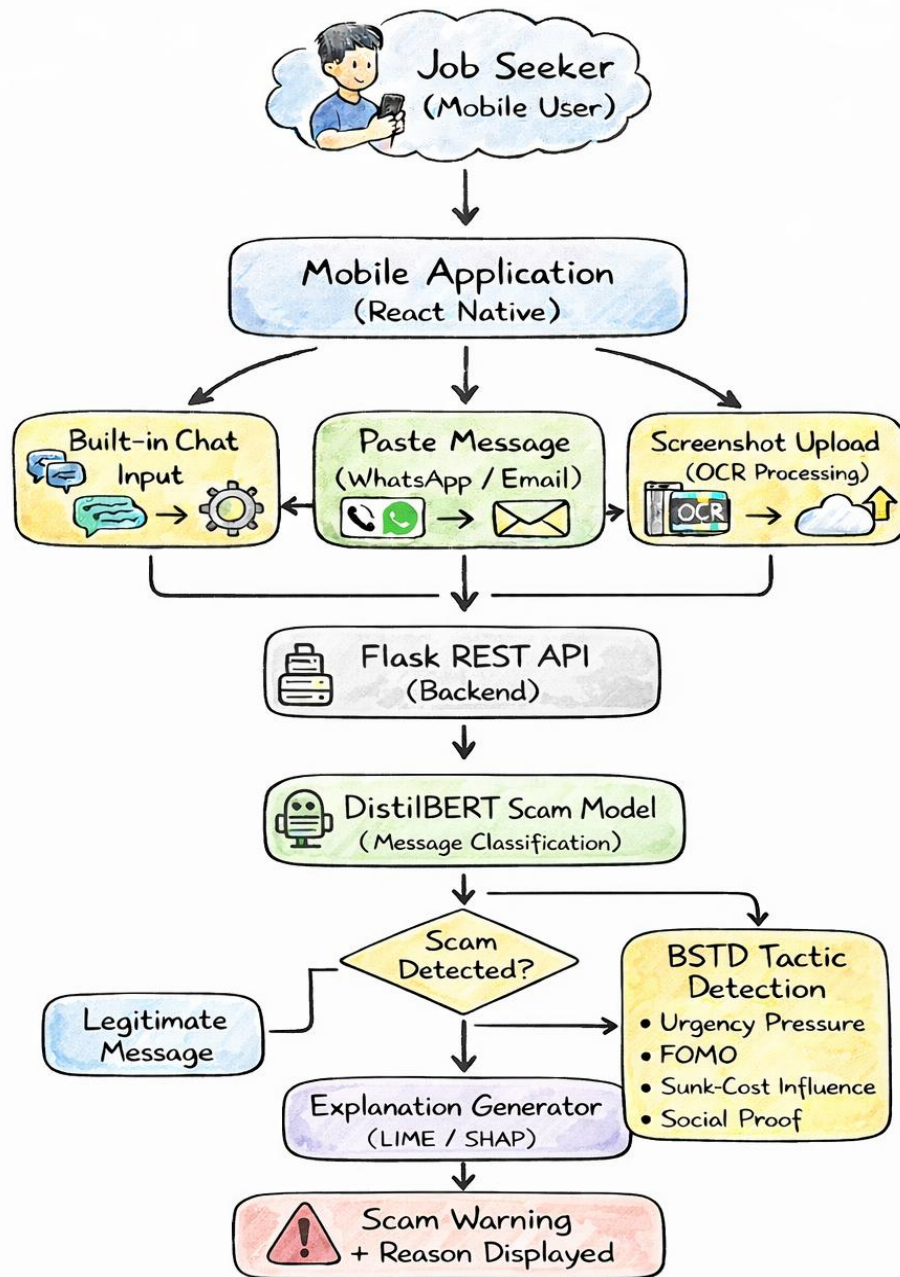
4.1 Overview of Proposed Solution

The proposed system is a mobile-based scam communication detection platform that analyses recruiter–candidate messages in real time to identify fraudulent content and psychological manipulation tactics. The primary function of the system is to protect job seekers during the active communication phase of recruitment fraud — the stage where financial loss and identity theft actually occur.

The system operates through three input modes. Mode 1 is the core function — a built-in chat platform where recruiters and candidates communicate directly inside the application, and every message is analysed in real time as the conversation takes place. Mode 2 allows users to paste or type messages received from external platforms such as WhatsApp, Telegram, SMS, LinkedIn, or email for instant analysis. Mode 3 allows users to upload a screenshot of any conversation, which is processed through an OCR pipeline to extract text before analysis.

The system is structured across three layers — a React Native mobile application, a Flask REST API backend, and an AI engine. This component operates as Component 3 within the four-component AI Ecosystem for Fake Job Scam Detection and Fraud-Aware Job Recommendation developed at SLIIT under CoEAI.

4.2 System Diagram



Proposed Recruitment Scam Communication
Detection System Architecture

Figure 2 : Proposed BSTD System Architecture

4.3 Individual Component and Its Parts

Mobile Application — React Native Provides the user interface for all three input modes. Mode 1 hosts the built-in recruiter–candidate chat. Mode 2 provides a text input screen for external messages. Mode 3 provides a screenshot upload screen. Results including scam or legitimate classification, tactic label, and plain-language warning are displayed to the user after each analysis

Flask REST API Acts as the communication layer between the mobile application and the AI engine. It receives input from the mobile app, routes it to the appropriate model, and returns the classification result and explanation to the user within the target response time of under 2 seconds.

TF-IDF and Logistic Regression — A baseline scam classification model using TF-IDF feature extraction and Logistic Regression will be trained and evaluated during the research phase. This model is not used in the live application but establishes the performance benchmark against which the fine-tuned DistilBERT model is compared to demonstrate measurable improvement.

Fine-Tuned DistilBERT — Primary Classifier The main classification model performs two tasks simultaneously — binary scam classification (scam or legitimate) and multi-label tactic classification identifying which of the four psychological manipulation tactics from Anagha et al. [2] is present in the message.

BSTD Tactic Classifier Identifies which specific psychological manipulation tactic the scammer is deploying — urgency pressure, FOMO, sunk-cost influence, social proof, or none. This directly operationalises the theoretical behavioural findings of Anagha et al. [2] into a working real-time detection system.

OCR Pipeline OpenCV pre-processes screenshot images through denoising and adaptive thresholding. Tesseract then extracts conversation text from the processed image. The extracted text is passed directly into the DistilBERT classification pipeline, enabling platform-agnostic detection without requiring API access from any messaging platform.

Explanation Engine Maps the detected tactic label to a plain-language warning template displayed to the user. For example, a message classified as urgency pressure generates the warning: *"This recruiter is creating artificial time pressure. Legitimate employers do not impose same-day deadlines."*

4.4 Component Interactions and Data Flow

1. User sends or receives a message through the built-in chat platform, or inputs an external message or screenshot via the mobile application
2. For Mode 3, the screenshot is passed through the OCR pipeline — OpenCV pre-processes the image and Tesseract extracts the text
3. The extracted or typed text is sent to the Flask REST API
4. The API passes the text to the fine-tuned DistilBERT model for binary scam classification — Scam or Legitimate
5. If classified as scam, the BSTD tactic classifier identifies which specific manipulation tactic is present — urgency pressure, FOMO, sunk-cost influence, or social proof
6. The Explanation Engine generates a plain-language warning mapped to the detected tactic
7. The Flask API returns the result to the mobile application
8. The classification result, tactic label, and plain-language warning are displayed to the user in real time

4.5 Scalability and Security Considerations

All communication between the mobile application and Flask API will be encrypted using HTTPS to protect message content in transit. No user message content will be permanently stored on the server after the classification result is returned. User data will be anonymised at collection and stored using synthetic ID codes in compliance with PDPA 2022. The system architecture is designed to handle multiple concurrent requests and can be scaled using cloud infrastructure as user demand grows.

5. USER REQUIREMENTS

5.1 Stakeholders

Table 2- Stakeholder Identification

Stakeholder	Description
Job Seekers	Primary users who use the mobile application to detect psychological manipulation tactics in recruiter messages and receive plain-language warnings
Recruiters	Users who communicate with candidates through the built-in chat platform within the application
Employment Platforms	Platforms such as ikman.lk and TopJobs.lk that may integrate the BSTD detection API to protect their users from recruitment fraud
CCID Sri Lanka	Law enforcement division that benefits from reduced online recruitment fraud complaints through automated detection

5.2 Functional Requirements

FR1: The system shall provide a built-in chat platform where recruiters and candidates can communicate, and every message shall be analysed in real time for scam detection.

FR2: The system shall allow users to paste or type an external message received from any platform and receive an instant scam classification result.

FR3: The system shall allow users to upload a screenshot of a conversation, extract the text using the OCR pipeline, and return a scam classification result.

FR4: The system shall classify each message as Scam or Legitimate using the fine-tuned DistilBERT model.

FR5: The system shall identify which psychological manipulation tactic is present in a scam message — urgency pressure, FOMO, sunk-cost influence, social proof, or none — using the BSTD tactic classifier.

FR6: The system shall generate a plain-language warning mapped to the detected tactic and display it to the user immediately after classification.

FR7: The system shall support message analysis across multiple platforms including WhatsApp, Telegram, SMS, LinkedIn, and email through Mode 2 and Mode 3 input.

FR8: The system shall allow users to view their previous analysis history within the mobile application.

5.3 Non-Functional Requirements

Performance: The system shall return a classification result within 2 seconds of receiving a message input under normal operating conditions.

Usability: The system shall achieve a System Usability Scale (SUS) score of 70 or above, evaluated through a user study with a minimum of 20 participants.

Security: All communication between the mobile application and Flask REST API shall be encrypted using HTTPS to protect message content in transit.

Privacy: No user message content shall be permanently stored on the server after the classification result is returned. All participant data shall be anonymised in compliance with PDPA No. 9 of 2022.

Reliability: The system shall be available and operational during all evaluation and user study sessions without unplanned downtime.

Scalability: The system architecture shall be capable of handling multiple concurrent user requests without significant degradation in response time

6. COMMERCIALISATION PLAN

6.1 Target Market and Customer Persona

The primary target market is job seekers aged 18–35 in Sri Lanka who actively use online employment platforms and mobile messaging applications. This demographic is most vulnerable to recruitment fraud, particularly recent graduates and first-time job seekers seeking employment through platforms such as ikman.lk and TopJobs.lk. The secondary target market is employment platforms themselves, which require fraud detection capabilities to protect their users and maintain platform credibility and trust.

6.2 Value Proposition

The proposed system is the first to detect psychological manipulation tactics in recruiter–candidate conversations in real time, providing job seekers with plain-language warnings that are immediately actionable. Unlike existing solutions that only flag fraudulent job advertisements, the proposed system protects users at the point where financial harm actually occurs — during active communication with scammers across any messaging platform.

6.3 Cost Estimation

The total estimated cost for developing the system is broken down as follows:

- **AWS EC2 / Render (Flask API + DistilBERT model serving): LKR 4,500/month** — Cloud hosting for the Flask REST API backend and DistilBERT inference engine during evaluation and user study phase.
- **MongoDB Atlas (data storage): LKR 3,000/month** — Cloud-hosted MongoDB Atlas instance for storing anonymised session data and analysis history.
- **Google Colab Pro (model training): LKR 2,500/month** — GPU-accelerated environment for DistilBERT fine-tuning and TF-IDF baseline training.
- **Play Store developer account: LKR 7,500 one-time** — Google Play Store registration fee for deploying the React Native mobile application for user study participants.
- **Annotation tool / labelling platform: LKR 1,500/month** — Online annotation platform for two independent annotators to label scam messages with binary and tactic labels.
- **Printing and consent forms: LKR 700 one-time** — Printed participant information sheets and consent forms for dataset collection and user study.
- **Total Monthly Cost: LKR 11,500/month**
- **Total One-time Cost: LKR 8,200**

6.4 Revenue Model and Pricing Strategy

The proposed system will be launched as a free mobile application to maximise adoption among Sri Lankan job seekers. A premium subscription tier at LKR 299 per month will provide full tactic detection history, priority analysis, and advanced warnings. As a future commercialisation opportunity, employment platforms such as ikman.lk and TopJobs.lk could license the detection API to embed scam protection directly into their platforms, generating B2B revenue potential beyond the initial consumer application.

6.5 Competitive Advantage

Factor	Advantage
Novelty	First and only system to detect psychological manipulation tactics in recruitment conversations in real time
Multi-platform	Works across WhatsApp, Telegram, SMS, LinkedIn, and email via text input and screenshot
Explainability	Plain-language warnings designed for non-technical job seekers
Local relevance	Designed specifically for the Sri Lankan job market and CCID-reported fraud patterns
No platform API needed	OCR pipeline enables detection without integration with any messaging platform

Table 3 Competitive Advantage

6.6 Intellectual Property Considerations

The proposed framework, tactic classification taxonomy, and annotated scam conversation dataset represent original intellectual contributions developed at SLIIT under CoEAI. The system will be submitted for provisional patent consideration through SLIIT's research commercialisation office. All open-source components used — DistilBERT, HuggingFace, Tesseract, and OpenCV — are used under their respective permissive licences.

7. Budget and justification

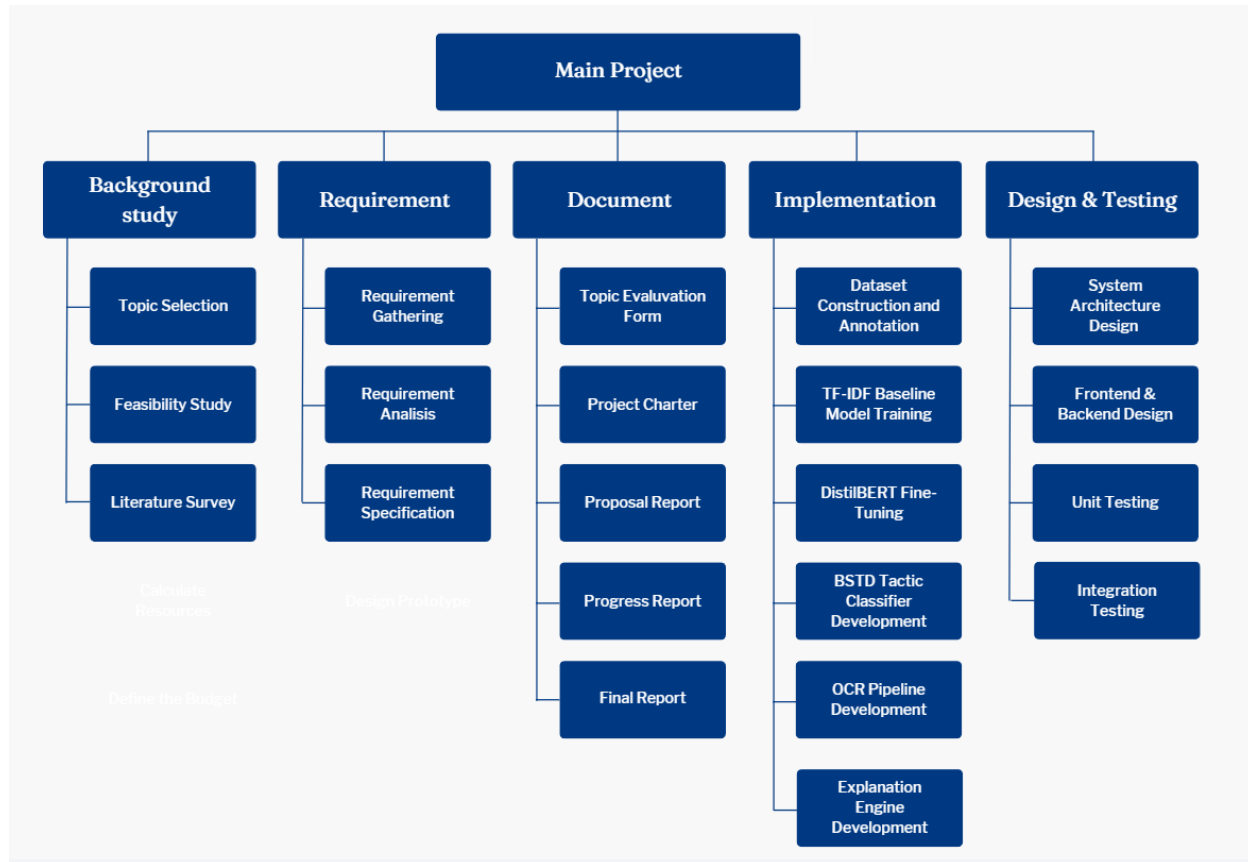
This research has been designed to minimise costs by selecting open-source tools and utilising existing hardware. All software tools including Python, HuggingFace, DistilBERT, Tesseract, OpenCV, Flask, and React Native are free and open-source, requiring no software licensing costs. Development and model training will be conducted on existing personal equipment, eliminating hardware costs. The primary recurring expenses are internet charges for model training and API communication, cloud hosting for the Flask REST API during evaluation, travel for participant recruitment and supervisor meetings, and printing for consent forms and report submission.

Table 4 : Budget plan per month

Component	Amount (USD)	Amount (LKR)
Travelling fee	3.00	1000.00
Internet Charges	15.00	5100.00
Printing and Papers	2.00	700.00
Hosting the application	9.00	3150.00
Total	29.00	9950.00

8. Work Breakdown Structure

Figure 3: Work Breakdown Structure



9. Gantt Chart

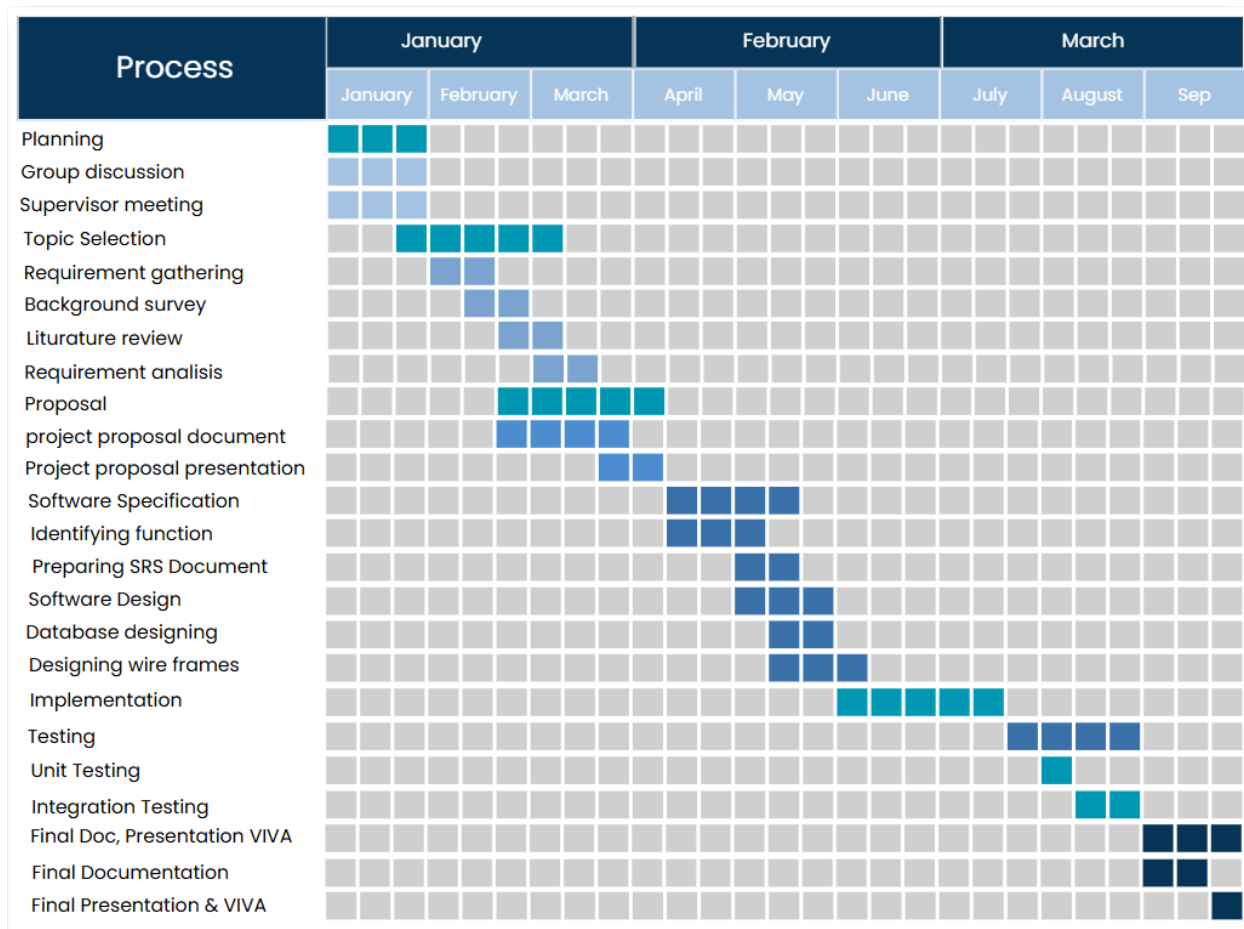


Figure 4 : Gantt Chart

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11 .Appendices

APPENDIX A — SURVEY INSTRUMENTS

A.1 Survey Overview

A preliminary survey was conducted between 2 March 2026 and 9 March 2026 to validate the research problem and assess the need for the proposed system. A total of 78 responses were collected from job seekers, undergraduate students, and working professionals in Sri Lanka.

The questions include

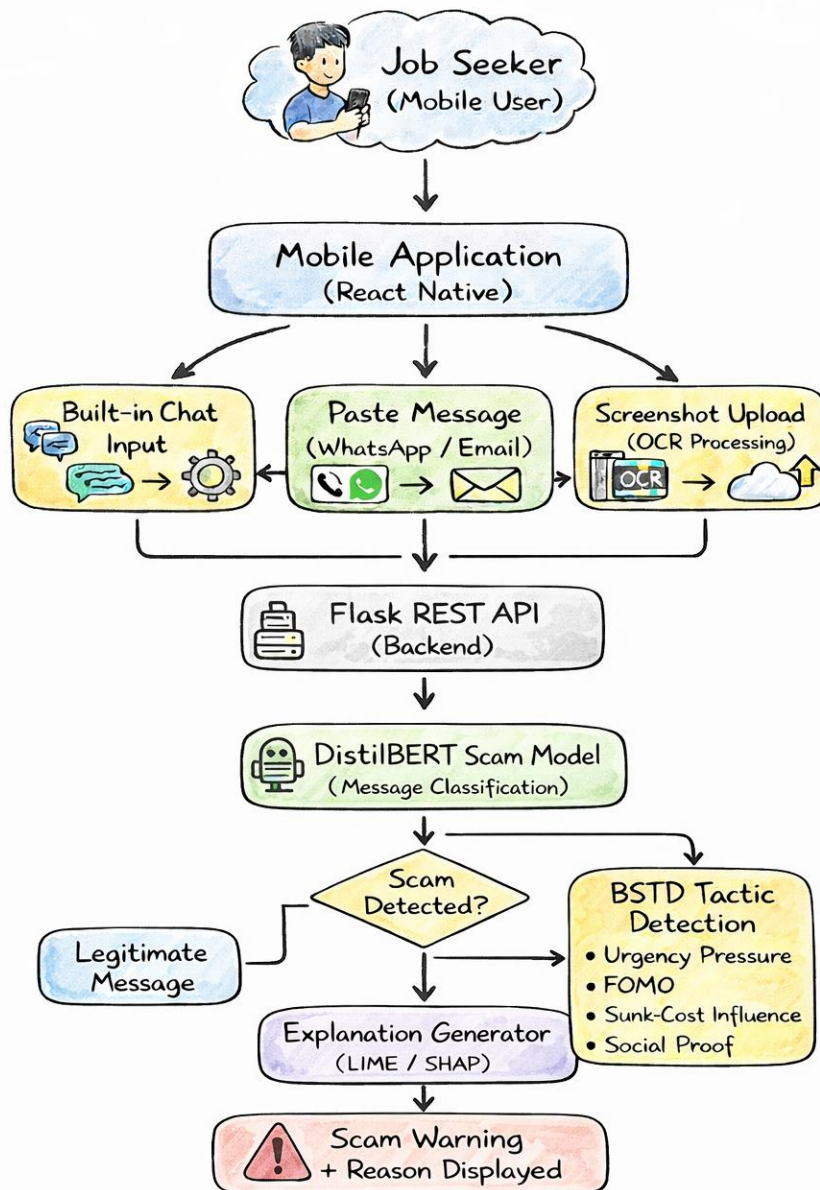
- What is your group age?
- What is your current status?
- Have you ever searched for jobs online?
- Which platforms do you used to search for jobs?
- Have your ever seen fake or suspicious job advertisements online?
- What characteristics make a job advertisement look suspicious?
- Do you usually check the company details applying for a job?
- Which company-related factors increase your trust in a job opportunity?
- Have you ever received suspicious messages from someone claiming to be a recruiter?
- Through which platforms did you receive suspicious recruiter messages?
- What type of suspicious behavior did you notice in those messages?
- Have you ever been asked to pay money for a job opportunity?
- How confident are you in identifying fake job offers or recruitment scams?
- Do you think recruitment scams are increasing on online platforms?
- Would you use a system that automatically detects scam job messages or suspicious recruiters?
- Which features would help you detect job scams more effectively?
- Do you think AI technology can help reduce recruitment scams and protect job seekers?
- In your opinion, what is the biggest risk when applying for jobs online?

A.2 Key Survey Findings

Finding	Result
Total responses	78
Respondents who have seen suspicious job advertisements	85%
Respondents who have received suspicious recruiter messages	68%
Most common suspicious behaviour reported	Asking for money or processing fee
Most common platform for suspicious messages	WhatsApp, LinkedIn, Telegram
Respondents who believe recruitment scams are increasing	82%
Respondents who would use an AI scam detection system	58% said Yes or Maybe
Most requested feature	Explanation of why a message is suspicious

Table 5 Key Survey Findings

APPENDIX B — Details Diagrams



Proposed Recruitment Scam Communication
Detection System Architecture

Figure 5: Proposed BSTD System Architecture

APPENDIX C — PRELIMINARY UI MOCKUPS

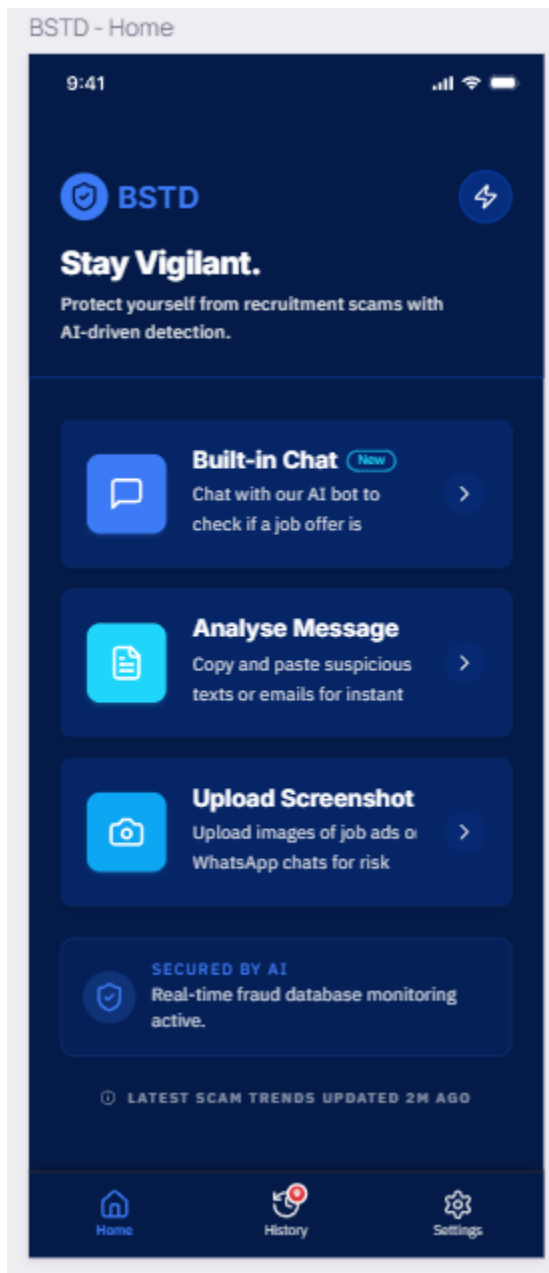


Figure 6: UI Mockup — Home Screen

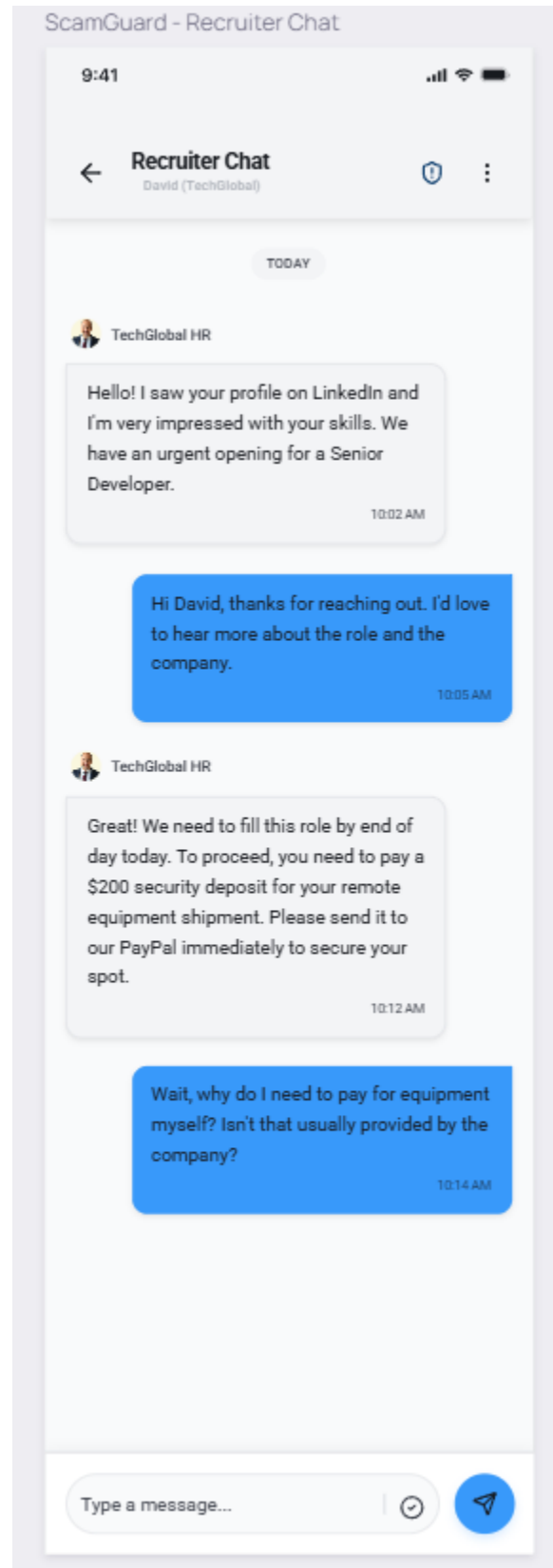


Figure 7: UI Mockup — Chat Screen

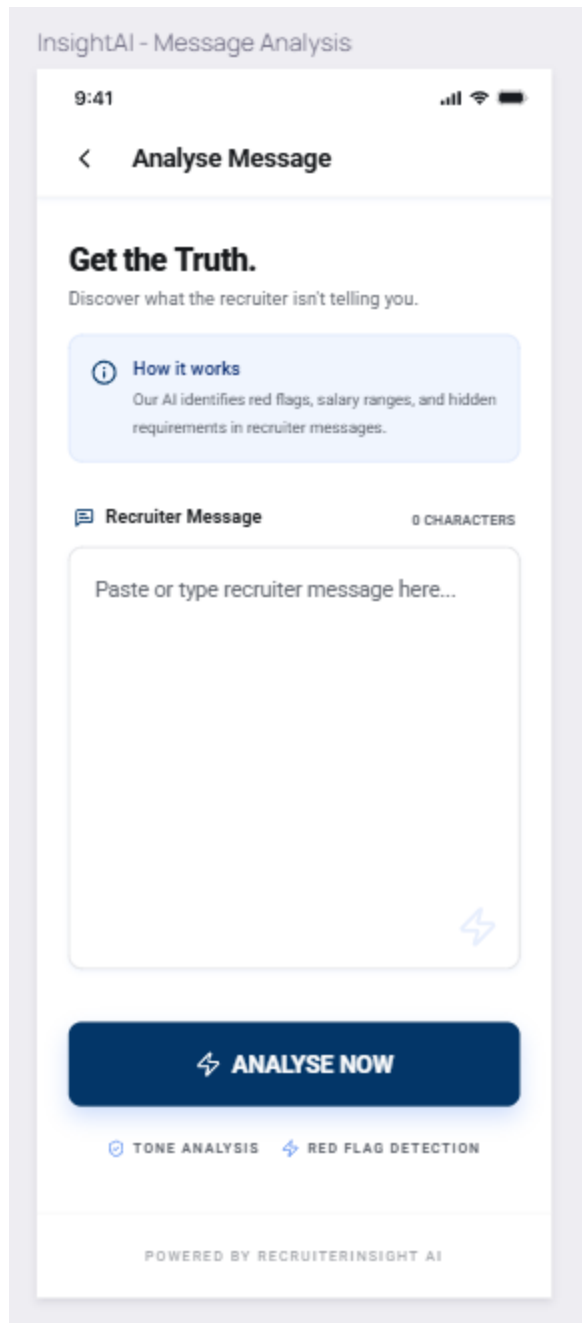


Figure 9: UI Mockup — Paste Message Screen

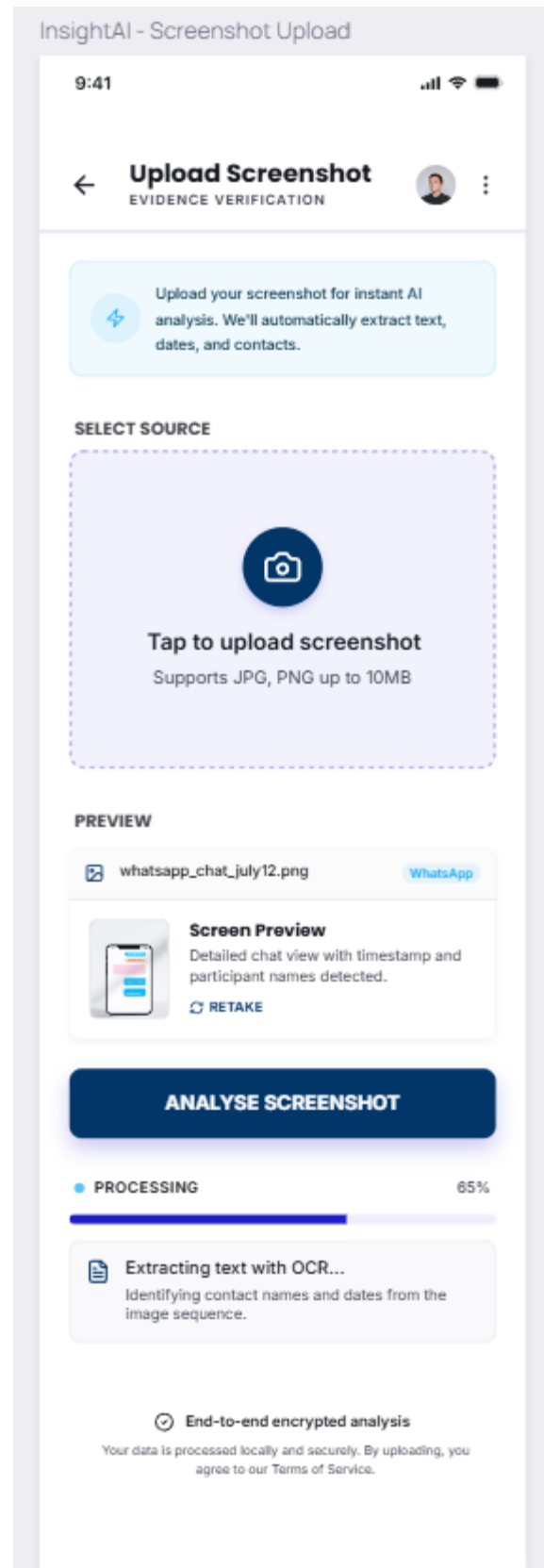


Figure 8: UI Mockup — Screenshot Upload Screen

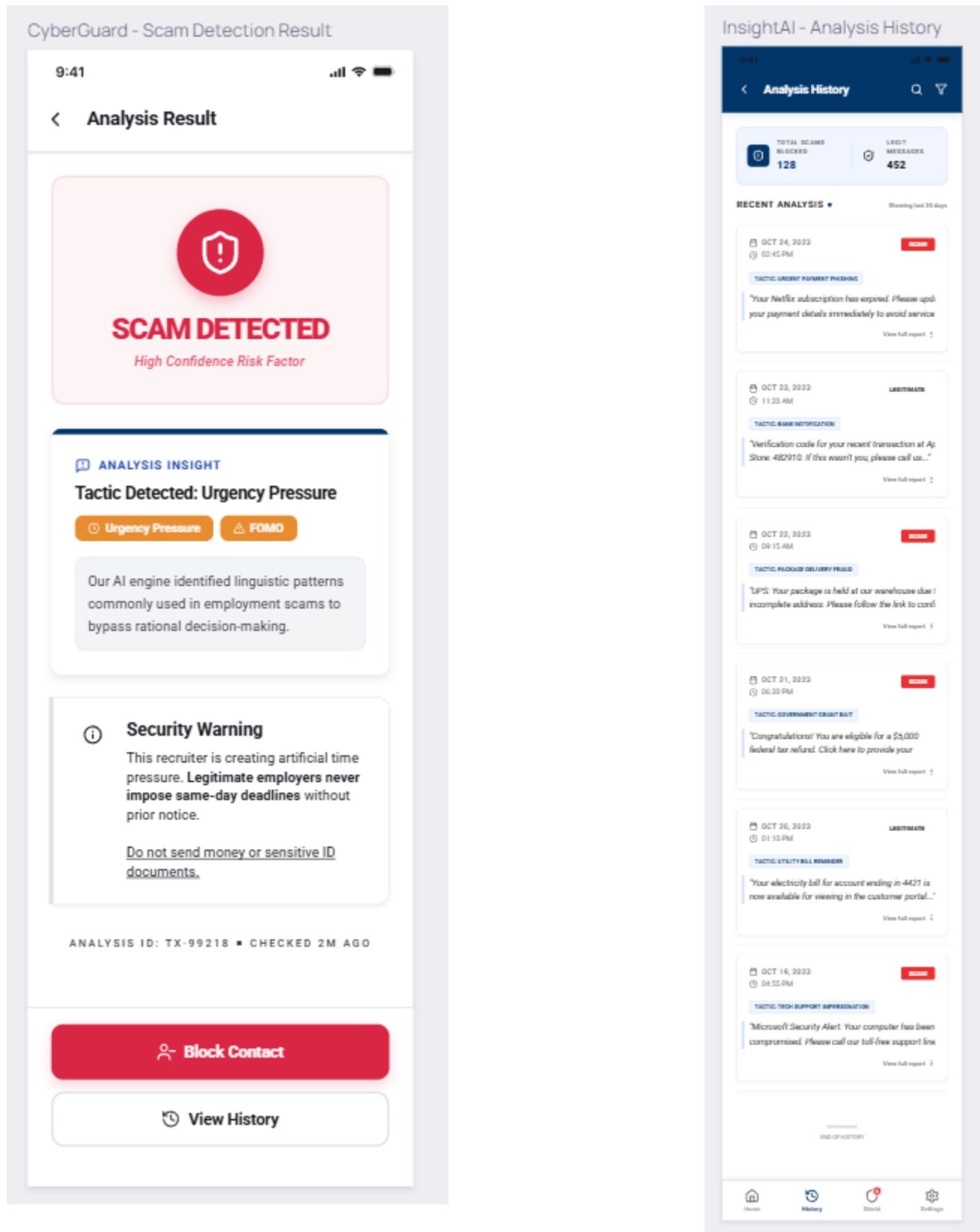


Figure 10: UI Mockup — Result Screen

APPENDIX D — RISK ANALYSIS

The possible dangers of the proposed system are listed in this appendix along with various mitigating techniques.

Risk	Description	Mitigation Strategy
Dataset unavailability	Anansi dataset access may not be approved by authors	Collect custom dataset from participants as primary source
Model performance	DistilBERT may not achieve target F1 score of 0.90	Experiment with hyperparameters and augment training data
OCR inaccuracy	OCR pipeline may produce low accuracy on real screenshots	Apply additional OpenCV pre-processing and test on diverse samples
Annotation disagreement	Annotators may not reach Cohen's Kappa of 0.80	Conduct calibration sessions before full annotation begins
Data privacy breach	Personal data may be exposed during data collection	Immediate anonymisation and PDPA No. 9 of 2022 compliance

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